

IDS FUNDAMENTALS:

- IDS stands for intrusion detection system. It is a system that monitors your network/system and alerts you if something suspicious happens.

IDS cannot block attack, it can only detect them and send alerts.

- Types of IDS on the basis of deployment mode:

a) Network Intrusion Detection system (NIDS): NIDS solutions are used to detect potential malicious activity within the whole network by monitoring traffic in and out in a network.

b) Host Intrusion Detection System (HIDS): HIDS solutions are downloaded on every single laptop in a network because they are used to detect potential security threats associated with that particular host. But HIDS can be challenging to manage in large network because they are resource intensive and require management on each host.

- Types of IDS based on Detection modes:

a) Signature-based IDS: Each attack has its unique pattern known as signature. These signatures are saved in IDS database are used to prevent from future attacks with the same patterns. However, signature-based IDS can only detect attacks with the pattern already stored in their database so if a new pattern attack happens it cannot detect it.

b) Anomaly based IDS: This type of IDS first learns the normal

behaviour of the network or system and then if anything except the normal happens it points it to be malicious. This IDS may generate a lot of false positive alert. But we can reduce them by manually defining the normal behaviour.

- c) ^{IDS} Hybrid IDS: A hybrid ^{IDS} uses both detection methods signature-based IDS and anomaly based IDS to leverage strength of each approach. If a attack pattern whose signature is already saved ~~from~~ in database then it uses signature base IDS for its detection. If its a new pattern attack then it use

signature based IDS can detect threats quickly but other IDS takes time to process. Signature-based IDS can be good for small threat surface but Anomaly based and hybrid are the best for modern day attacks.

• Smart?

Smart is a IDS solutions. It is ~~used~~ ~~as a~~ ~~Network~~ Intrusion detection system. It detects malicious activities on the network and raise alerts. It uses both ^(main) signature based and ^(limited) anomaly based detections to identify threats. Smart comes with built-in detection rules, which you can also disable and create your own custom rules.

Modes of Smart?

- i) Packet Sniffer mode: This mode of smart only allows you to read and display the packets without performing any analysis. It doesn't relate directly to IDS capabilities but can be used to monitoring traffic and troubleshooting.

b) Packet logging mode: This mode of smart allows you to log the network traffic as PCAP (standard packet capture format) file. It logs the network's traffic but if any detection is made by IDS mode it can also log that, logging mode itself doesn't detect anything.

c) Network Intrusion Detection system: This is the primary mode of smart that monitors network's traffic and raises alert on detection of any malicious activity by the help of attack patterns saved as Signature.

• Smart usage:

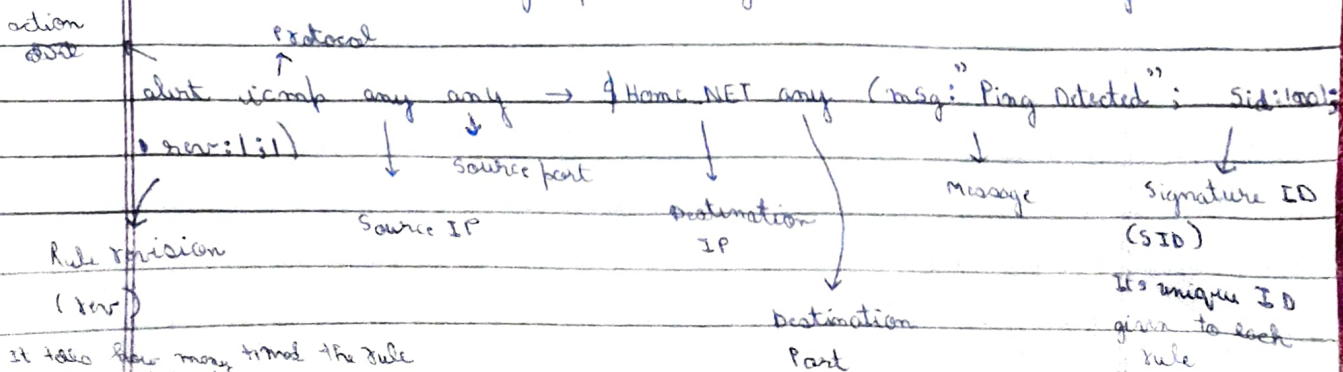
Smart has some built in rules files, configuration file and other files. These are in ~~smart~~ etc/smart directory. You can specify rules. Smart's main file is its configuration file where you specify which rule file to enable and which network range to monitor.

Rules are stored in rule folder.

• Rule format:

There is a particular format in which rules are written or defined in smart.

Example: This rule will detect a ICMP packets (usually when you ping a host) coming from any IP and port and reaching the some network to any port. It generates a alert "Ping Detected".



Home-NET: In this example is a variable whose value is already saved in configuration file which the network range.

Rule Metadata: Rule's metadata is defined at the end of the rule inside the brackets.

• Rule creation:

For this we have to paste the rule inside "local.rules" ~~directory~~ file present in `etc/snort/rules/local.rules`.

How to start snort tool to detect any intrusions defined in the rule file?

```
sudo snort -q -l /var/log/snort -i lo -A console -c /etc/snort/snort.config
```

- q → ²quite mode less output
- l /var/log/snort → saves log in this folder or output in this folder
- i lo → monitor interface
- A console → show alerts on screen
- c /etc/snort/snort.config → use config file

• How to identify which mode is snort running in?

Sniffer mode	No config, Just <code>snort -v</code>
Packet logger mode	-l + no local rules
IDS mode	-c snort.config

How are local rules applied?

We wrote our rules in `etc/snort/rules/local.rules` and in the config file there is a line ^{include} `$rule_path/local.rules` so when we load config file it loads local rules and rules are applied.

• Running Snort on PCAP files?

There are cases when our network traffic is logged and these are logged in Standard Packet Capture format ^{" "}PCAP. Snort can also be used to detect intrusions on these PCAP files containing historical network traffic.

Command for this :

```
sudo snort -q -l /var/log/snort -r Task.pcap -A console -c /etc/snort/snort.config
```

Reads packet from this file instead of live network.